Intro to Comp Architecture

Quiz #4

Question #1:

Response Time:

response time is the total (latency) time required to complete one task or program

Response time = Time for one jobs to finish

Throughput:

throughput is the number of tasks completed in a given amount of time

$$\text{Throughput} = \frac{\text{no of jobs completed}}{\text{unit time}}$$

e.g A webserver is upgraded with a faster network card (1Gbps → 10Gbps). throughput increases because more requests can be handled per second, but if each request still requires same CPU processing time, the average response time per request remains unchanged.

Question #2:

(a) Performance Ratio

$x = 20$, $y = 15s$

$$\text{Performance Ratio} = \frac{\text{Time } x}{\text{Time } y} = \frac{20}{15} = 1.333$$

y is 1.33 times more faster than x

(b) Improvement

$$\text{Improvement} = \frac{20-15}{20} \times 100\% = 25\%$$

y is 25% faster than x.

Question #3:

(a) Iron Law

CPU time = Instruction count × CPI × clock period.

Given: Instruction = 2×10^9

Machine A: CPI = 1.6, clock period = 0.4 ns = 0.4×10^{-9}

$T_A = 2 \times 10^9 \times 1.6 \times 0.4 \times 10^{-9}$

$= 2 \times 1.6 \times 0.4$

$\boxed{T_A = 1.28\,s}$

Machine B: CPI: 2.0, clock period = 0.25 ns = 0.25×10^{-9}.

$T_B = 2 \times 10^9 \times 2.0 \times 0.25 \times 10^{-9} = 2 \times 2.0 \times 0.25 = 1.0$ second.

(b) $\frac{T_A}{T_B} = \frac{1.28}{1} = 1.28\,s$

Machine B is 1.28 times faster than Machine A

Question #4:

- Before the floating point hardware:
  - One floating point operation required using:
    - ↳ 50 integer instruction
    - ↳ Each instruction = 1 cycle

So instruction count = 50 and cycles = 50.

Given clock rate: 1 MHz

$$MIPS = \frac{\text{Instruction count}}{\text{Execution time} \times 10^6}$$

MIPS $\frac{50}{50} = 1$

1 MIPS

- After floating point hardware:
  - same operation uses: 1 floating point instruction
    - ↳ takes 2 cycles

So instruction count: 1

cycles: 2

MIPS $= \frac{1}{2} = 0.5$

$\boxed{= 0.5\ \text{MIPS}}$

MIPS decreased from 1 to 0'5 but execution time improved dramatically:

- Old execution: 50 cycles
- New execution: 2 cycles

Thus lower the MIPS faster the actual performance.

MIPS can be misleading because it depends on instruction count not actual work completed.

Assignment #4

FA23-BCS-1??
D... Fatima

Question #1:

(a) Weighted Mean $= w_1x_1 + w_2x_2$

Machine 1 : $(0.7)(8) + (0.3)(4.0)$

$= 5.6 + 12$

$\boxed{M_1 = 17.6s}$

Mach 2: $(0.7 \times 4) + (0.3 \times 70)$

$= 2.8 + 21 = 23.8s$

$\boxed{M_2 = 23.38s}$

(b) Overall Performance:

$17.6 < 23.8$

Machine 1 gives better overall performance

(c) Un weighted Arithmetic Mean

Machine 1 $\frac{8+40}{2} = 24s$

Machine 2 : $\frac{4+70}{2} = 37s$

They would think machine 1 is faster

Question #2:

(a) Total Distance / Total time

Time for 1st segment :

$t_1 = \frac{20}{60} = 1/3\ h$

Time for 2nd segment:

$t_2 = \frac{20}{120} = \frac{1}{6}h$

Total time $= \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$ h

Total distance $= 20+20 = 40$ km

Total speed $= \frac{40}{(1/3)}$ $\boxed{= 80 km/h}$

(b) Harmonic Mean

for equal distance, harmonic mean is used:

$$H = \frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(120)}{60+120} = \frac{14400}{18} = 80$$

$\boxed{H = 80 km/h}$

∴ harmonic mean is used because speed ~~rate~~ is a rate and equal distances are travelled.

Question # 3:

(a) statement of Amdahl's law

Overall speed up =>

$$S = \frac{1}{(1-f) + (f/s)}$$

f = fraction improved

s = speed factor

(b) Overall speed up

Non vectorizable = 25%

vectorizable = 75%

speed up = 8

$$S = \frac{1}{0.25 + \frac{0.75}{8}}$$

$$= \frac{1}{0.25 + 0.09375}$$

$\boxed{S = 2.91}$

(c) Maximum possible speedup

If $s \to \infty$

$$\frac{0.75}{\infty} = 0$$

$$S = \frac{1}{0.25} = 4$$

Limit exists because non vectorizable part 25% cannot be speed up, it becomes the bottle neck.

Question #4:

(i) Arithmetic Mean of Times

It is correct when measuring execution times for same workload across machines (consistent with total times).

(ii) Harmonic mean of rates:

correct for averaging rates (km/h, transaction/sec) when measuring same distance or task count, because total time = sum of (distance/rate).

(iii) Geometric mean of ratios

correct for normalizing performance relative to a reference machine especially for benchmarks of different types because it avoids dependence on the reference choice.

Why arithmetic mean?

different reference of machines can produce contradictory conclusions:

e.g. machine A appears faster than B using one reference

. But slower from other reference

Arithmetic mean does not properly handle ratios.

Geometric mean avoids inconsistency and gives fair comparison results